

Architect

14 Years Experience

ABOUT ME

Driven Software Architect with 14 years of experience in designing, developing, and deploying scalable software and system-level solutions across diverse domains. Proficient in Python, Go, Rust, and C++, with hands-on expertise in both application and system programming — including device communication, multi threading, and performance optimisation. Experienced in building data-driven architectures using Spark, Kafka, and Delta Lake, and developing robust APIs and micro-services with FastAPI, Flask, and Beego. Combines strong technical depth with a passion for crafting reliable, maintainable, and high-performance systems that align with business goals.

CURRENT COMPANY

Apr 2018 - Present

Valueadd Softtech

Responsibilities

- Continuously explore, evaluate, and adopt emerging technologies to strengthen solution capabilities and system scalability.
- Design and architect robust, high-performance software and data solutions aligned with business objectives.
- Actively contribute to development by writing clean, efficient, and maintainable code across multiple languages and frameworks.
- Mentor and guide team members through design reviews, code reviews, and implementation best practices.
- Collaborate with cross-functional teams to ensure alignment between architecture, development, and deployment strategies.
- Oversee the full life cycle of solution delivery — from design and development to deployment and post-release optimisation.
- Foster a culture of continuous improvement in architecture, performance, and engineering excellence.

SKILLS

- | | | | |
|-----------------|-----------|---------|--------------|
| • Python | • Go Lang | • C++ | • Rust |
| • FastAPI | • Flask | • Beego | • MongoDB |
| • MS SQL Server | • PySpark | • Kafka | • Delta Lake |
| • Kedro | • Pandas | • Numpy | • DuckDB |

TOOLS

- | | | | |
|-----------|----------|-------------|----------|
| • Airflow | • Docker | • UV | • Poetry |
| • Git | • Jira | • Cursor AI | |

PROJECTS WORKED ON

1. Neural Vision

Duration of Project: March 2025 - Present

Technology Used: Rust, Dioxus, OpenCV, Tokio

Tools used: GitHub

Team Size: 5

Project Description:

Neural Vision is a multi-threaded, cross-platform application written in Rust designed to empower users with image recognition and computer vision capabilities. It provides a robust and user-friendly platform for training patented hardware neural networks, performing real-time object recognition and classification, and managing complex computer vision projects.

Role in the Project:

As the Technical Lead for the Neural Vision project, I implemented solutions based on the architectural design, guided team members through development, and took ownership of the entire code base to ensure adherence to best practices and coding standards. I designed and built the multi-threaded core framework in Rust, establishing a robust inter-thread communication mechanism for efficient concurrency and performance. My role involved exploring and integrating modern Rust frameworks such as Tokio, Crossbeam, rusb etc., developing modules for direct hardware communication, and driving continuous technical improvement across the project.

2. ImpactCraft

Duration of Project: 1 Month

Technology Used: Python, Kedro, Poetry, MongoDB

Tools used: BitBucket, Docker, Jenkins

Team Size: 1

Project Description:

ImpactCraft is a comprehensive solution that empowers Customer Success (CS) teams to streamline CS processes, improve organizational performance, and enhance customer retention through data-driven insights and automation.

Role in the Project:

I explored and evaluated various orchestration tools and data workflow frameworks to identify the most suitable architecture, and set up the project structure from scratch. I designed and implemented data pipelines using Kedro to collect, clean, and transform data from multiple sources, storing the processed data in MongoDB for downstream machine learning workflows. I also collaborated with the DevOps team to design and implement efficient deployment strategies, ensuring scalability, and reliable delivery across environments. Additionally, I developed and maintained reusable, modular components to support common functionalities across multiple data pipelines and clients, improving maintainability and consistency across the system. Also guided members from client team through development, and took ownership of the entire code base to ensure adherence to best practices and coding standards.

3. IWT Analytics

Duration of Project: 8 Months

Technology Used: Python, PySpark, Kafka, Delta Lake, FastAPI, SQL Server

Tools used: GitHub, Grafana, Jenkins

Team Size: 6

Project Description:

IWT Analytics Portal is a mining-focused application designed to monitor and manage mine operations in real-time. It provides live dashboards and reports across modules like Location, Production and Atmosphere. The Location module tracks miner positions, check-ins, section timings, and under/above ground movement. The Atmosphere module monitors gas levels (CO, H2, N2), absolute pressure, and triggers alarms for unsafe conditions, while also providing hourly and daily min/max/average statistics.

Role in the Project:

As the Lead for the IWT Analytics Portal project, I provided technical leadership and direction for developing the processing engine for mining operations monitoring platform. I designed the database schema and Delta Lake data models, and implemented Kafka-Spark PySpark streaming jobs and batch data processing pipelines to handle high-frequency operational data efficiently. I created supporting tools and utilities to generate synthetic data, analyse real data and emulators to push data to system in absence of real data. I optimised processing workflows to reduce resource consumption and improve system performance. Guided the team in developing scalable APIs to serve processed data to the web portal, ensuring timely and accurate insights across modules such as Location and Production. Additionally, I led and mentored a cross-functional team, driving adherence to best practices, coding standards, and delivery timelines.

4. OCR - Publications

Duration of Project: 24 Months

Technology Used: Python, Flask, Pandas, BeautifulSoup, MongoDB

Tools used: BitBucket, Docker, Jenkins

Team Size: 6

Project Description:

OCR – Publications is a data-driven platform that provides U.S. trade regulations and export licensing requirements in an easy-to-query and visualize format. It simplifies the interpretation of complex regulatory data, enabling users to quickly understand compliance obligations and make informed business decisions.

Role in the Project:

As the Lead Developer for the OCR – Publications project, I designed and developed the entire solution from scratch to process and visualize U.S. regulatory data extracted from eCFR (Electronic Code of Federal Regulations). I implemented ETL pipelines to extract, transform, and store hierarchical data, designed logic to flatten and reconstruct complex hierarchies for visualization, and introduced a time-travel feature to analyze historical data changes over time. In addition to core development, I built RESTful APIs using Flask, ensuring efficient data access and seamless integration with visualization layers. I collaborated closely with the team to maintain code quality, resolve issues, and deliver a scalable, maintainable, and high-performance solution.

OTHER PROJECTS

Worked on several impactful projects across domains including electric vehicle charging, recruitment automation, and emergency response systems. As Tech Lead for ConnectMyEV, developed REST APIs and embedded web services in C++ for smart EV charging systems deployed on Raspberry Pi platforms. Led the RapidHyre project to build scalable microservices using Go (Beego) and Python (Django) for automated candidate screening through AI-powered telephony and analytics. At Rolta, contributed to multiple mission-critical systems such as CAD Desktop, MDT, Communication Server, and Messaging Server, developing high-performance C++ applications with Boost ASIO and VC++ for real-time incident management, GPS tracking, and message relaying in emergency response and public safety environments.

WORK HISTORY

**Valueadd Softtech &
Systems Pvt. Ltd.**

7.5 Years | Apr 2018 - Present | Architect

Rolta India Ltd.

6.6 Years | Sep 2011 - Mar 2018 | Sr. Software Engineer

PORTFOLIO/ GITHUB LINKS

<https://github.com/nileshs-vast>